Lesson plan · for the teacher

Poisoned data

AI security · Level 3 · Lesson 1

Time
40 min

Materials
Fruit, leaves or stones of two kinds (or a printed table of their measurements); A ruler or a kitchen scale; Paper, pencils and the board

Learning goals

- Explain how poisoned data and backdoors spoil a model.
- Find examples far from their group's median.
- Plan checks before new data goes into training.

Activity: Find the poison

1. Warm-up (5 min): Ask: if you had to teach a machine to tell mangoes from limes, what examples would you give it? Agree that it can only learn from the examples it is given.
2. Measure (10 min): Groups weigh or measure 10 to 15 items of two kinds and call out each value and its label. Write them in two columns on the board.
3. Poison (5 min): While pupils look away, swap three labels on the board. Tell them some data has been poisoned, but not how much.
4. Hunt (15 min): Each group sorts the values of each kind, finds the middle value (the median) and flags any value further than half the median away. Compare with the real swaps: which were caught, and which honest items were flagged too?
5. Wrap-up (5 min): Discuss why a flagged item is checked before it is removed, and agree two rules for collecting class data. Then give the exit ticket.

No scale? Use lengths of leaves or sticks measured with a ruler, or print a table of 15 measurements. Choose two kinds whose values don't overlap much, so the swapped labels stand out.

Discussion

- Why could the poison change the model even though nobody touched it?
- Which honest items were flagged, and why?
- What rules would keep our class data clean?

Exit ticket

- Write one way poison gets into data, and one check that would catch it.

Try it at school tomorrow
Play “Find the poison”: pupils measure fruit, leaves or stones and label each one, you secretly swap three labels, and groups find the median of each kind to hunt for the swapped ones.

Practice questions · for you

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1 **Pupils add 100 new photos and the score on your clean test set drops from 97% to 80%. What now?**

- ◯ Keep the photos: more data is always better
- ◯ Stop, check the new photos, and train without the suspicious ones
- ◯ Change the test set until the score goes back up

2 **Where can poisoned examples get into a school's training data? Choose all that apply.**

- ☐ A clean test set nobody can change
- ☐ Measurements typed in by pupils
- ☐ Data scraped from websites anyone can edit
- ☐ Labels added by strangers online

3 **What does this program print?**

```
ws = [12, 11, 13, 90, 12]
m = sorted(ws)[len(ws) // 2]
print(m, sum(ws) / len(ws))
```

◯ 90 12.0 ◯ 13 27.6 ◯ 12 27.6

4 **Finish far(ws): it returns the weights that sit further from the median than half the median. The two fruit lists should give [40] and [190].**

```
def far(ws):
    m = sorted(ws)[len(ws) // 2]
    out = []
    for w in ws:
        # keep w if it is further from m
        # than half of m
        pass
    return out

print(far([210, 195, 40, 205]))
print(far([38, 42, 190]))
```

Write your program:

5 **Every example the odd-one-out check flags is poisoned.**

◯ True ◯ False

✎ **Where does data that your pupils or your school use with AI come from? Where could a wrong or planted example slip in, and how would you notice?**

Check your answers

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1 **Stop, check the new photos, and train without the suspicious ones**

A sudden drop on a clean test set is the warning sign.

2 **Data scraped from websites anyone can edit; Labels added by strangers online; Measurements typed in by pupils**

Open websites, strangers' labels and pupils' entries can all carry wrong or planted examples.

3 **12 27.6**

Sorted, the list is 11, 12, 12, 13, 90, so the median is 12.

4

```
def far(ws):
    m = sorted(ws)[len(ws) // 2]
    out = []
    for w in ws:
        if abs(w - m) > m / 2:
            out.append(w)
    return out

print(far([210, 195, 40, 205]))
print(far([38, 42, 190]))
```

The median is the middle value of the sorted weights; a weight further than half of it away is far from its kind, so it gets checked by a person before anything is removed.

5 **False**

The check flags anything unusual, including honest examples near the border, like a really small mango.